AMAL JYOTHI COLLEGE OF ENGINEERING
(AUTONOMOUS),KANJIRAPPALLY

PYTHON PROJECT REPORT - COLLEGE EVENT REMINDER

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ABSTRACT

The College Event Reminder System is a Python-based website developed to help students manage and keep track of important college events. In many academic environments, students often miss events such as seminars, workshops, cultural programs, and examinations due to poor communication or lack of reminders. This system provides a simple and efficient solution for organizing and monitoring upcoming events.

The application allows users to view and admin to add, view, edit, and delete event details such as event name, date, time, category, and description. All event information is stored in a JSON file, ensuring that the data remains saved even after the program is closed. The system also includes a live countdown timer that displays the remaining time until the next upcoming event.

The user interface is designed using modern Python GUI libraries such as CustomTkinter and Tkinter Calendar, which provide a visually appealing and user-friendly experience. By combining event management, countdown tracking, and notification features, the College Event Reminder System helps students stay informed and better organized with their academic and extracurricular activities.

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DESCRIPTION:

In colleges and universities, many events such as seminars, workshops, cultural programs, sports activities, and examinations are conducted regularly. Students often find it difficult to remember the dates and times of all these events. Important announcements are usually shared through notice boards, messages, or social media groups, but students may overlook them or forget about them later. As a result, they may miss important academic or extracurricular activities.

Currently, there is usually no simple and centralized system where students can easily check and track all upcoming college events in one place. Without a proper system, students must rely on memory, handwritten notes, or scattered online messages, which can be inconvenient and unreliable. This can lead to confusion, missed opportunities, and reduced participation in college activities.

To address this problem, a **College Event Reminder Web Suite** was developed. The system allows users to store and manage event details such as event name, date, time, category, and description in a structured format. Users can view upcoming events, manage event information, and track the next upcoming event through a simple interface.

The system is currently designed as a **web-based solution with supporting Python components**, making it accessible and easy to manage. In the future, the same system can be extended into a **full mobile or desktop application** with additional features such as automatic reminders, user accounts, and real-time event updates.

By providing a centralized event management platform, the system helps students stay organized and ensures that important college events are not missed.

1.TOOLS AND LIBRARIES USED

Libraries and Tools Used

Python

Python is the main programming language used in this project. It is used to implement the core logic of the system such as adding events, storing data, retrieving event details, and calculating upcoming events.

CustomTkinter

CustomTkinter is a modern graphical user interface (GUI) library built on top of Tkinter. It is used to design a visually appealing interface with components such as buttons, frames, labels, input fields, and tabs.

Tkinter

Tkinter is the standard GUI library available in Python. It is used in the basic prototype of the system to create windows, buttons, entry fields, and lists for displaying events.

TkCalendar

TkCalendar provides a calendar widget that allows users to select dates easily. This helps users choose event dates from a calendar instead of manually typing them.

Plyer

Plyer is a cross-platform library used to generate desktop notifications. It is used in the system to remind users about events that are scheduled for the current day.

JSON

JSON is used as the data storage format in the system. Event details such as name, date, time, category, and description are stored in a file called **events.json**.

HTML

HTML is used to structure the web interface of the project. It defines the layout of the web pages and displays event information.

CSS

CSS is used to style the web pages and improve the visual appearance of the system.

JavaScript

JavaScript is used to add interactivity to the web interface, such as dynamically displaying event data.

VSCODE

Visual Studio Code is the development environment used to write, edit, and run the project files.

2.TOOLS AND SERVICES

Tools and Services Used

- **Visual Studio Code (VS Code)**

Visual Studio Code was used as the primary development environment for writing, editing, and organizing the project files. It provides features such as syntax highlighting, debugging support, and extension integration which help in efficient coding.

- **Python Interpreter**

The Python interpreter is used to execute the Python scripts of the system. It runs the backend logic responsible for managing events, processing data, and performing reminder calculations.

- **Web Browser**

A web browser is used to view and test the web interface of the system. It allows the user to interact with the interface and verify that the layout and functionality work correctly.

- **JSON File Storage**

The project uses a JSON file (events.json) as a lightweight data storage method. It stores event information such as event name, date, time, category, and description, allowing the system to retrieve the data whenever the program runs.

- **Desktop Notification Service (Plyer)**

The Plyer library provides a cross-platform notification service. It allows the system to send desktop notifications to remind users about events happening on the current day.

- **Calendar Date Selection Service (TkCalendar)**

The TkCalendar service provides a graphical calendar widget that enables users to select event dates easily, improving accuracy and usability.

- **CustomTkinter UI Framework**

CustomTkinter is used as a modern user interface framework for building the graphical components of the system such as forms, buttons, labels, and dashboard panels.

3.SYSTEM DESIGN

The College Event Reminder Web Suite is designed to manage and display college events in an organized manner. The system follows a simple structure where the user interacts with the interface to add or view events, and the backend processes and stores the event data.

The overall design of the system consists of three main components: the user interface, the event processing logic, and the data storage module.

- **User Interface Layer**

This layer provides the interface through which users interact with the system. It includes input fields for entering event details such as event name, date, time, category, and description. It also displays the list of upcoming events and the next event reminder. The interface is designed to be simple and easy to use.

- **Event Processing Layer**

This layer handles the main functionality of the system. It processes user inputs, adds new events, updates existing events, deletes events, and calculates the next upcoming event. It also manages features such as countdown timers and event reminders.

- **Data Storage Layer**

All event data is stored in a JSON file called events.json. This file keeps the details of each event such as the event name, date, time, category, and description. The system reads from this file when loading events and writes to it whenever a new event is added or an existing event is modified.

4.WORK FLOW

1. The user opens the system interface.
2. The system loads previously stored events from the JSON file.
3. The user can add new events by entering event details.
4. The system saves the event data into the JSON file.
5. The system displays all upcoming events in the interface.
6. The program calculates the nearest upcoming event and displays the countdown timer.
7. If an event occurs on the current day, a reminder notification is generated.

This design ensures that the system remains simple, organized, and easy to extend in the future.

5. IMPLEMENTATION OF CODE

5.1. Backend Logic Implementation:

The backend logic of the College Event Reminder Web Suite is implemented using Python. The backend is responsible for handling event data, managing storage, and performing operations such as adding, updating, deleting, and retrieving events.

Event data is stored in a JSON file named **events.json**. When the program starts, the system reads this file and loads all previously saved events. Whenever a new event is added or an existing event is modified, the updated information is written back to the JSON file to ensure data persistence.

The backend also performs important processing tasks such as:

• Loading and saving event data using JSON functions
• Adding new events with details such as event name, date, time, category, and description
• Updating existing event details when edits are made
• Deleting events when they are no longer needed
• Sorting events based on date to display upcoming events first
• Identifying the next upcoming event and calculating the remaining time for the countdown timer

The system uses Python's **datetime module** to compare event dates with the current date and determine upcoming events. This allows the system to track and display the nearest upcoming event effectively.

Additionally, the backend integrates with the **Plyer notification library** to generate desktop notifications for events occurring on the current day.

5.2. GUI (Graphical User Interface) Implementation:

The graphical user interface of the system is designed using **CustomTkinter**, which is a modern extension of the Tkinter library. CustomTkinter allows the interface to have a clean and visually appealing design with modern UI components.

The GUI consists of two main sections:

• **Dashboard**
The dashboard displays the next upcoming event along with a live countdown timer showing the remaining days, hours, minutes, and seconds until the event begins. It also lists all upcoming events in a structured card layout.

• **Admin Panel**
The admin panel allows users to manage events. Users can add new events by entering details such as event name, date, time, category, and description. The admin panel also provides options to edit or delete existing events.

The interface includes various UI elements such as:

• Input fields for entering event details
• Buttons for adding, updating, or deleting events
• Calendar date picker using TkCalendar
• Scrollable frames for displaying event lists
• Labels and panels to organize information clearly

The GUI is designed to be user-friendly so that users can easily navigate the system and manage events without technical knowledge.

5.3. Multimedia and Asset Integration:

The system integrates several interface elements and visual assets to improve usability and appearance.

• **Icons and Emojis** are used within the interface to visually represent features such as events, reminders, editing, and deletion.
• **Color themes and styling** are applied using CustomTkinter to provide a modern dark-themed interface.
• **Calendar widget (TkCalendar)** is integrated to allow users to select event dates from a visual calendar instead of typing them manually.
• **Desktop notifications** are generated using the Plyer library to alert users when an event is scheduled for the current day.

These multimedia and visual elements enhance the overall user experience and make the system more interactive and visually engaging.

6. RESULT AND SCREENSHOTS

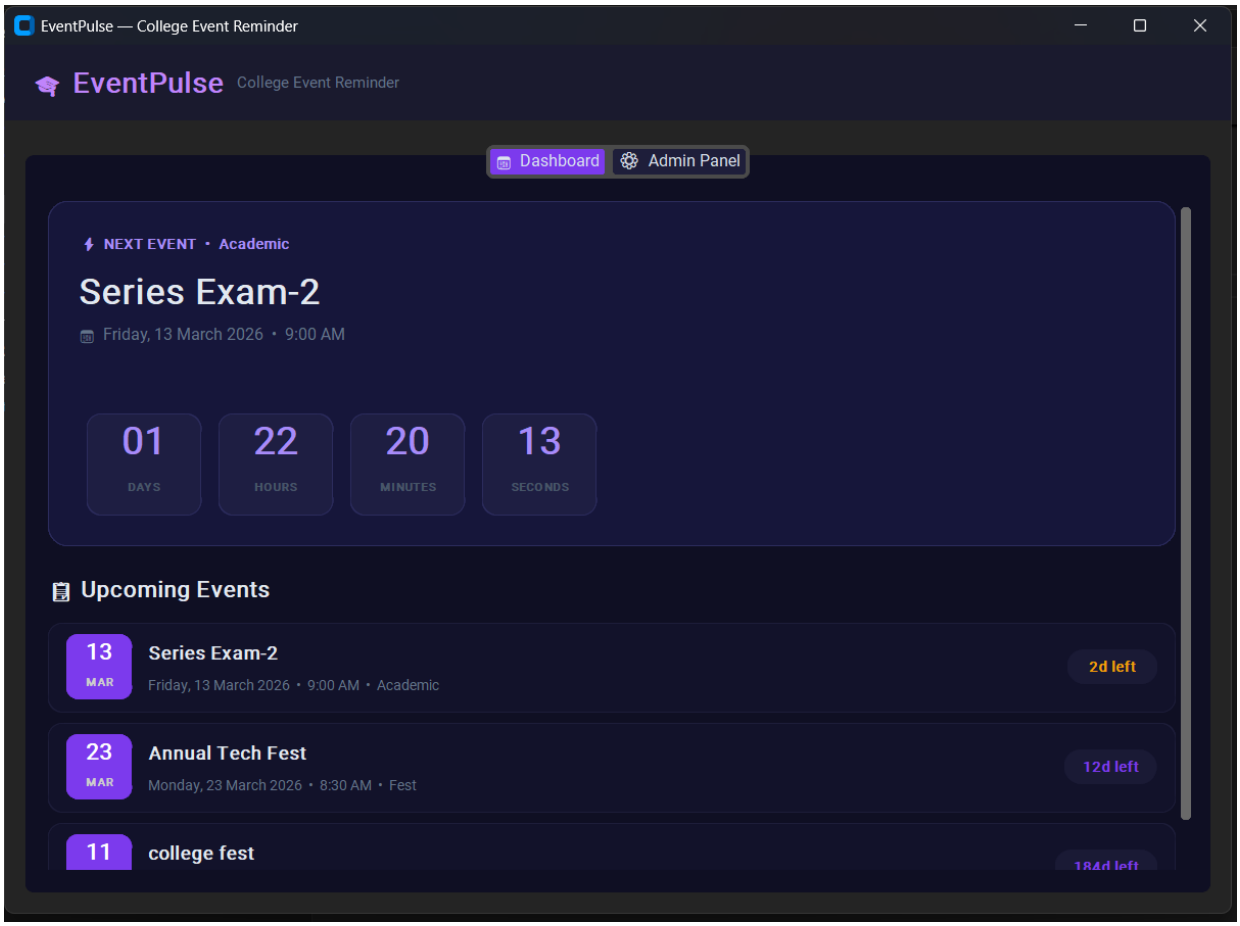
The College Event Reminder Web Suite was successfully developed and tested. The system is able to manage college events efficiently by allowing users to add, view, edit, and delete event information through a simple interface.

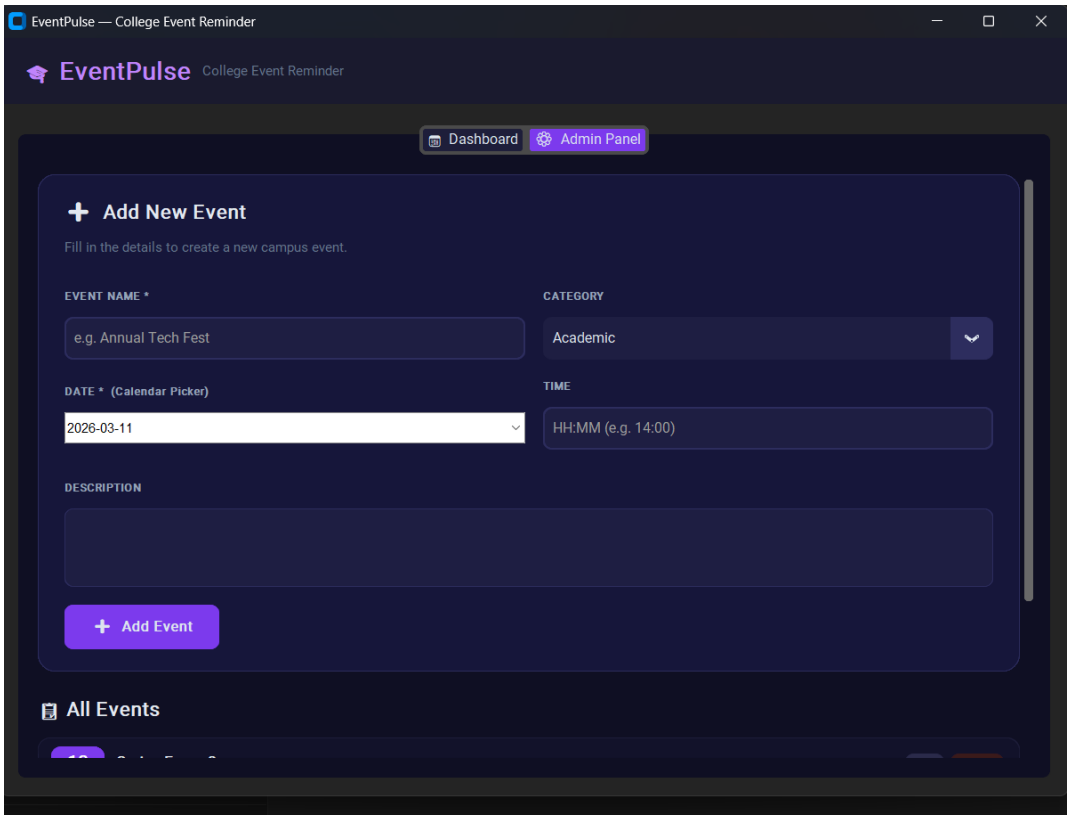
When the application is launched, the system loads previously stored event data from the **events.json** file. The dashboard displays the **next upcoming event** along with a **live countdown timer** showing the remaining days, hours, minutes, and seconds until the event occurs. This helps users quickly understand which event is approaching next.

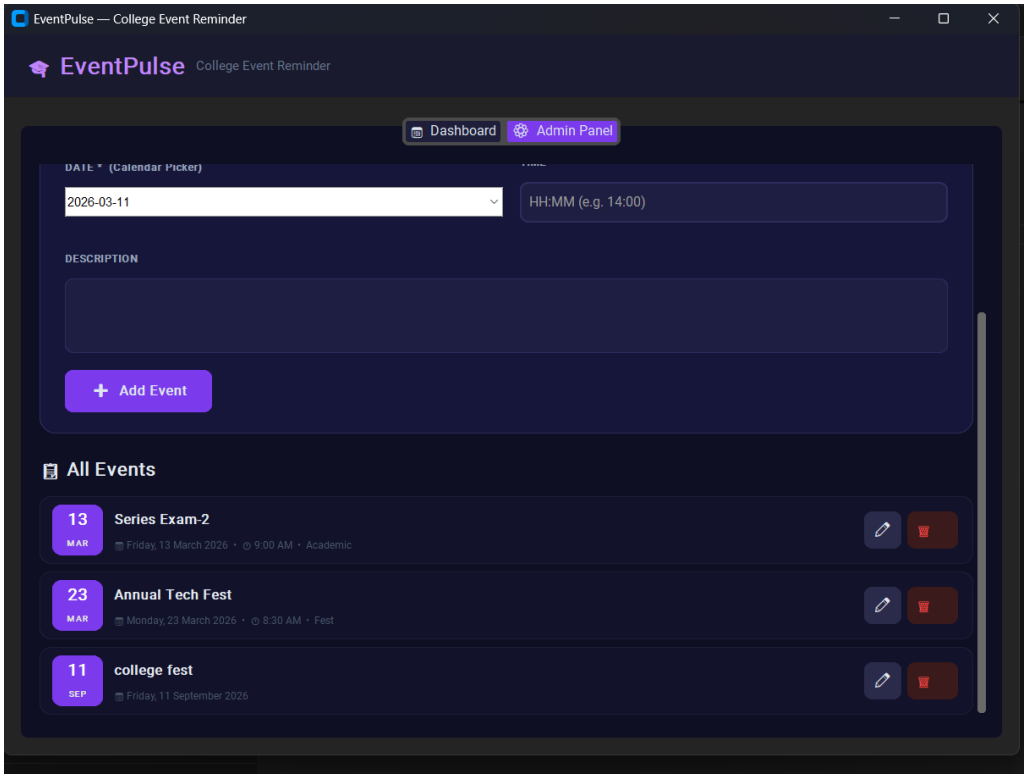
Users can open the **Admin Panel** to add new events by entering details such as the event name, date, time, category, and description. Once the event is added, it is immediately stored

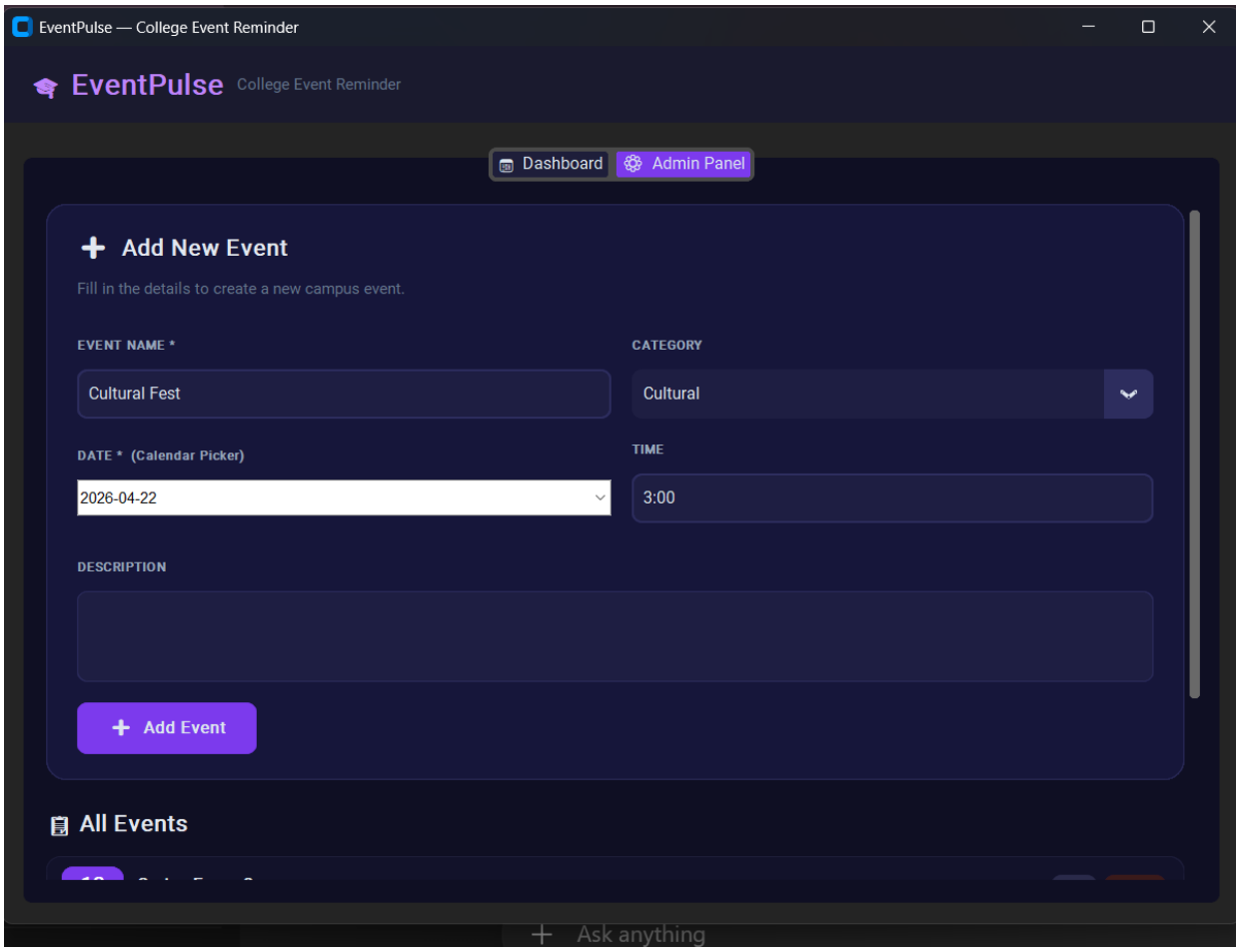
in the JSON file and displayed in the event list. The system also allows users to edit or delete events if any changes are required.

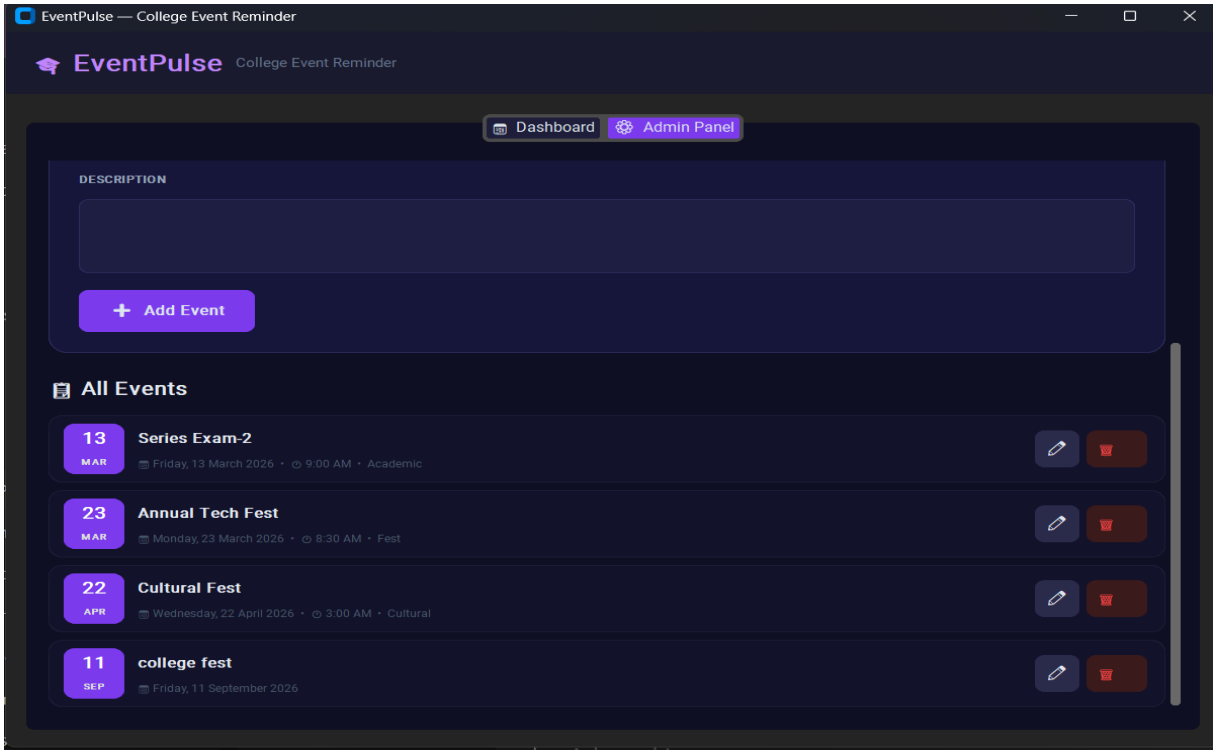
6.1.SCREENSHOT OF RESULT

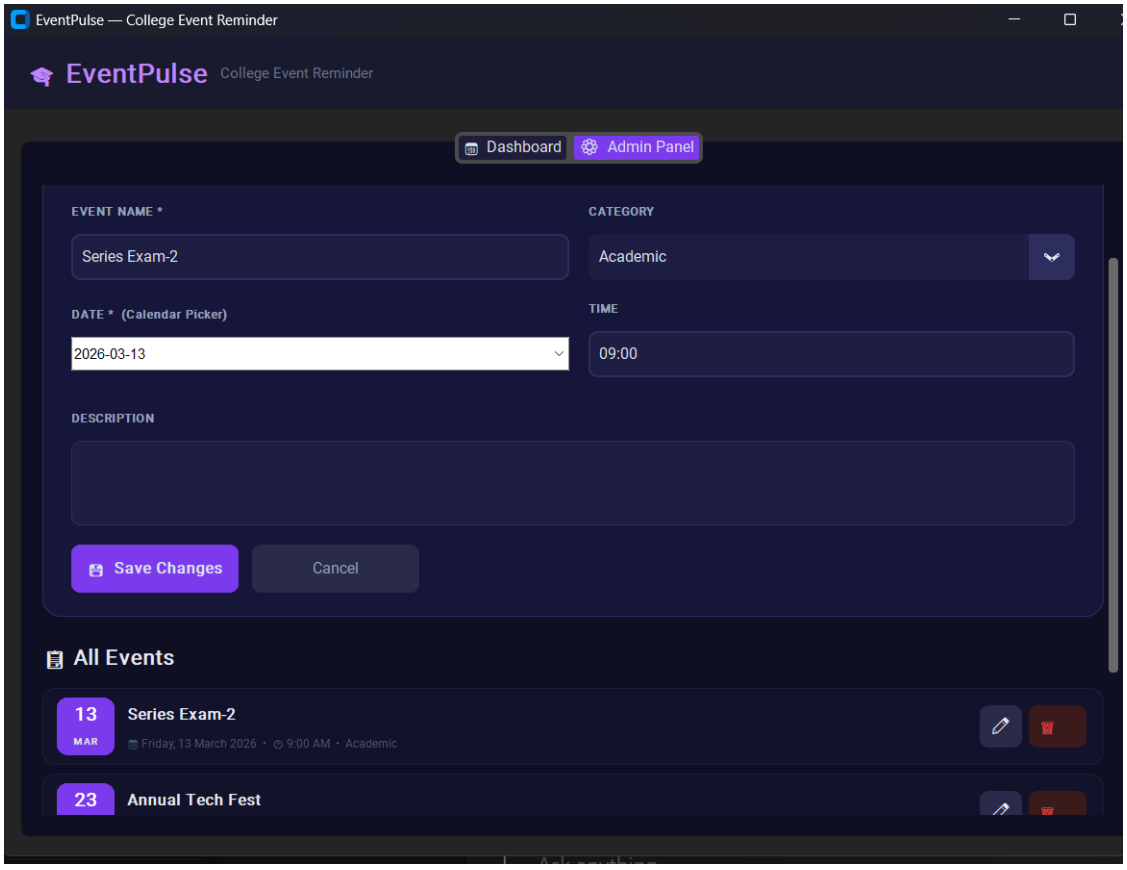


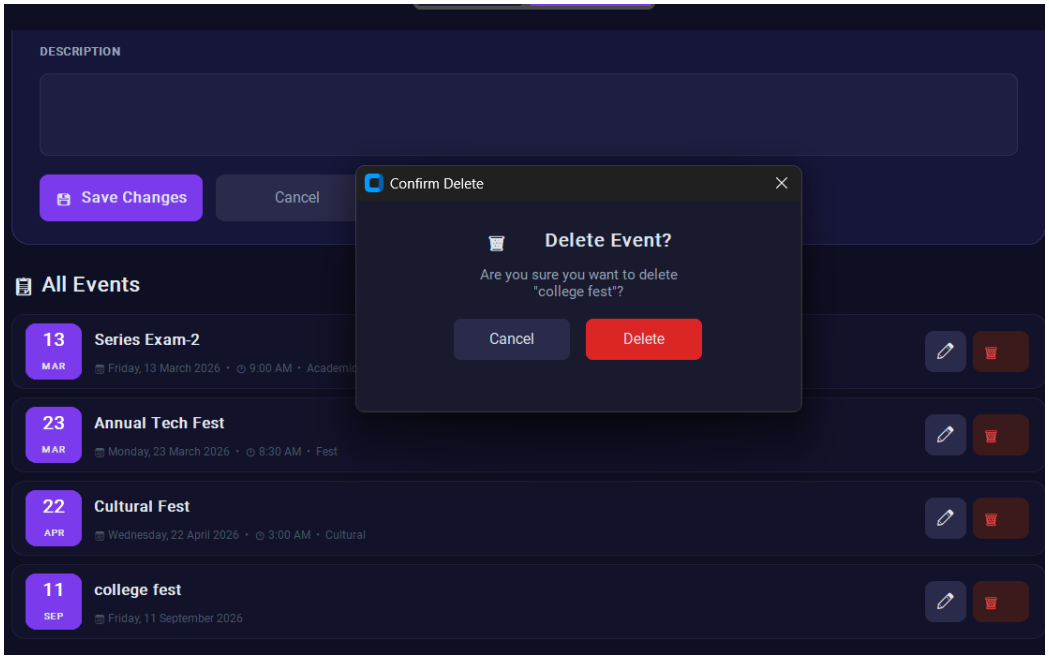

EventPulse — College Event Reminder
EventPulse
College Event Reminder
Dashboard
Admin Panel
Add New Event
Fill in the details to create a new campus event.
EVENT NAME *
e.g. Annual Tech Fest
CATEGORY
Academic
DATE * (Calendar Picker)
2026-03-11
TIME
HH:MM (e.g. 14:00)
DESCRIPTION
Add Event
All Events


EventPulse — College Event Reminder
EventPulse
College Event Reminder
Dashboard
Admin Panel
2026-03-11
HH:MM (e.g. 14:00)
DESCRIPTION
Add Event
All Events
13
MAR
Series Exam-2
Friday, 13 March 2026 • 9:00 AM • Academic
23
MAR
Annual Tech Fest
Monday, 23 March 2026 • 8:30 AM • Fest
11
SEP
college fest
Friday, 11 September 2026


EventPulse — College Event Reminder
EventPulse
College Event Reminder
Dashboard
Admin Panel
Add New Event
Fill in the details to create a new campus event.
EVENT NAME *
Cultural Fest
CATEGORY
Cultural
DATE * (Calendar Picker)
2026-04-22
TIME
3:00
DESCRIPTION
Add Event
All Events
Ask anything


EventPulse — College Event Reminder
EventPulse
College Event Reminder
Dashboard
Admin Panel
DESCRIPTION
Add Event
All Events
13 MAR
Series Exam-2
Friday, 13 March 2026 • 9:00 AM • Academic
23 MAR
Annual Tech Fest
Monday, 23 March 2026 • 8:30 AM • Fest
22 APR
Cultural Fest
Wednesday, 22 April 2026 • 3:00 AM • Cultural
11 SEP
college fest
Friday, 11 September 2026


EventPulse — College Event Reminder
EventPulse
College Event Reminder
Dashboard
Admin Panel
EVENT NAME *
Series Exam-2
CATEGORY
Academic
DATE * (Calendar Picker)
2026-03-13
TIME
09:00
DESCRIPTION
Save Changes
Cancel
All Events
13
MAR
Series Exam-2
Friday, 13 March 2026 • 9:00 AM • Academic
23
Annual Tech Fest


DESCRIPTION
Save Changes
Cancel
Confirm Delete
Delete Event?
Are you sure you want to delete "college fest"?
Cancel
Delete
All Events
13
MAR
Series Exam-2
Friday, 13 March 2026 • 9:00 AM • Academic
23
MAR
Annual Tech Fest
Monday, 23 March 2026 • 8:30 AM • Fest
22
APR
Cultural Fest
Wednesday, 22 April 2026 • 3:00 AM • Cultural
11
SEP
college fest
Friday, 11 September 2026

7.CONCLUSION AND FUTURE SCOPE

7.1. Conclusion:

The College Event Reminder Web Suite successfully demonstrates how event information can be organized and managed using a simple and efficient system. The project provides a centralized platform where users can add, view, edit, and delete college event details such as event name, date, time, category, and description.

The system stores event information in a JSON file, ensuring that all data remains saved and can be accessed whenever the application is launched. The dashboard displays upcoming events and highlights the next event with a live countdown timer, helping users easily track important dates. Additionally, the integration of desktop notifications ensures that users receive reminders for events occurring on the current day.

The project also demonstrates the effective use of Python libraries such as CustomTkinter, TkCalendar, and Plyer to build an interactive and user-friendly interface. Overall, the system provides a practical solution for organizing college events and helping students stay informed about upcoming activities.

7.2. Future Scope:

• The system can be expanded into a **fully functional mobile or desktop application** so that users can access event reminders from multiple devices.

• A **user login and authentication system** can be added to allow different users or departments to manage their own events.

• The project can be extended into a **complete web-based platform** where events can be updated online and accessed by students in real time.

• A **notification scheduling system** can be implemented to send reminders before the event time (for example, one day or one hour before the event).

• Additional features such as **event search, filtering by category, and calendar view** can be included to improve usability.

• Integration with **email or mobile notifications** can further improve the effectiveness of event reminders.

With these improvements, the system can evolve into a full-scale event management platform suitable for use across entire educational institutions.